CSE 323: Operating Systems Design

Section 3

Quiz 1

Total Points: 20 (4 Questions)

Allowed Time: 30 minutes

NAME:

ID:

Q1 **[5 Points]** The Operating System (OS) consists of multiple components. It "talks" to different hardware devices hooked up to your computer.

a. Which component of the OS directly controls hardware devices/resources? **[1 Point]**

- Kernel

b. What does it mean for the OS to be interrupt driven? **[2 Points]**

- OS mainly sits idle, Won't constantly check devices.
- When devices need attention, they issue interrupts

c. What is a Trap? How is it different from an Interrupt? **[2 Points]**

- Software generated interrupts due to errors, system calls.
- Interrupts originate from hardware devices.

Q2 **[5 Points]** The main memory is a crucial component of your computer system

a. What does it mean for your main memory to be volatile? **[1 Point]**

- Loses all content when power is cut.

b. Why does the CPU only load instructions from the main memory? Why is the main memory much smaller compared to secondary memory? **[2 Points]**

- Faster access time
- Expensive.

c. What is multiprogramming? How does the OS use multiprogramming to decrease CPU idle time? **[2 Points]**

- Bring multiple programs to main memory
- When one program is waiting, the OS schedules a different program to the CPU.

Q3 **[5 Points]** A process is represented by a PCB. A process cycles between many stages.

a. What is the full form of a "PCB" ? **[1 Point]**

Process Control Block

b. When does a process go from Running to Waiting? Is it a mandatory transition? **[2 Points]**

- Needs I/O to continue executing
- Mandatory as it cannot proceed without I/O

c. When does a process go from Running to Ready? Is it a mandatory transition? **[2 Points]**

- Time Slice expired
- Not mandatory as process could have continued if it was not pulled from the CPU.

Q4 **[5 Points]** A parent process creates a new child process that is a near-identical copy of itself. Consider a parent process (PID = 2) and a child process (PID = 4)

a. Which special function does it call to achieve this? **[1 Point]**

fork()

b. What is the value of the local variable pid of the parent process after the special function executes and returns? What is the pid value of the child process? **[2 Points]**

Parent → pid = 4

Child → pid = 0

c. Why is it hard to implement the Shared Memory architecture for Cooperating Processes? Why is it faster than Message Passing? **[2 Points]**

- Need to setup synchronization manually e.g. Producer/Consumer architecture
- Message Passing requires System Calls. Shared Memory does not.